Lesson 3: Shock Absorber Practical and Bar Charts

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students collect categoric data, record repeated measurements, calculate a mean, and present the results using a bar chart.

Assumptions

Students have been introduced to independent variables, dependent variables, continuous data, and categoric data in Lesson 2.

Practical title

Which material is the best shock absorber?

Investigation question

How does the type of material affect the bounce height of a ball?

Why this practical works well

This practical is fun, simple, and visual. Students drop a ball onto different materials and measure the bounce height. It naturally leads to categoric data and bar charts.

Learning objectives

By the end of the lesson, students should be able to:

1. Identify a categoric independent variable.
2. Collect repeated measurements.
3. Calculate a mean value.
4. Draw a results table with units.
5. Decide that a bar chart is appropriate.
6. Describe possible random and systematic errors.

Key vocabulary

Keyword	Student-friendly definition
Categoric data	Data sorted into groups or types
Trial	One repeat of a measurement
Mean	The average value
Random error	Unpredictable variation between repeats
Systematic error	A repeated error that shifts all results in the same way

Equipment

Per group:

- Marble, tennis ball, or small bouncy ball
- Meter ruler
- Clamp stand, if available
- Tray or box lid
- Different materials, for example sponge, cardboard, cloth, bubble wrap, rubber mat
- Exercise book or results table sheet

Safety notes

- Students should not throw the ball.
- The ball should be dropped vertically from a controlled height.
- Materials should be placed on a stable surface.
- Students should avoid crowding around the drop zone.

Starter

Review the rule:

> *If the independent variable is a category, use a bar chart.*

Ask:

- Is material type continuous or categoric?
- What graph should be used for this investigation?

Expected answer:

Material type is categoric, so a bar chart should be used.

Variables

Variable type	Example
Independent variable	Type of material
Dependent variable	Bounce height in cm
Control variables	Drop height, type of ball, release method, ruler position

Method

1. Choose 4 or 5 materials to test.
2. Place the first material on the bench or tray.
3. Hold the ball at the same drop height each time, for example 50 cm.
4. Drop the ball without pushing it.
5. Measure the bounce height.
6. Repeat the measurement 3 times for the same material.
7. Calculate the mean bounce height.
8. Repeat for the other materials.

Example results table

Material	Bounce height trial 1 / cm	Bounce height trial 2 / cm	Bounce height trial 3 / cm	Mean bounce height / cm
Sponge				
Cardboard				
Cloth				
Rubber mat				
Bubble wrap				

Data processing

Students draw a bar chart of material type against mean bounce height.

Bar chart expectations

A good bar chart should have:

1. A clear title.
2. Material type on the x-axis.
3. Mean bounce height on the y-axis.
4. Units on the y-axis.
5. Equal-width bars.
6. Gaps between bars.
7. A sensible scale.

Discussion questions

1. Which material was the best shock absorber?
2. Was the best shock absorber the material with the highest bounce or the lowest bounce?
3. Why did we repeat each measurement 3 times?
4. Did any result look unusual?
5. What could cause random errors in this experiment?
6. What could cause systematic errors in this experiment?

Error discussion

Error type	Simple meaning	Example in this practical
Random error	Unpredictable variation between repeats	Hard to judge the exact bounce height
Systematic error	An error that affects all results in the same way	Ruler starts from the wrong position

Plenary

Students complete the sentence:

> *I used a bar chart because the independent variable was ___________.*

Then ask them to add:

> *One possible random error was ___________.*

Possible prep

Students are given a results table from a similar categoric investigation and must draw a bar chart.

Teacher notes

The aim is not perfect data. The aim is for students to experience repeated readings, variation between trials, averages, and the logic of using a bar chart for categoric data.